Estimating Hourly Marginal Emissions in Interconnected Power Systems Using MRIO Accounting and Linear Model Trees

1st Noé Husser
dept. name of organization (of Aff.)
Twig
Copenhagen, Denmark
noe@twig.energy

2nd Liyang Han
dept. name of organization (of Aff.)
Twig
Copenhagen, Denmark
email address or ORCID

3rd Junesoo Shin
dept. name of organization (of Aff.)
Twig
Copenhagen, Denmark
email address or ORCID

***Abstract*—Marginal Emissions Factors (MEFs) can provide an operational signal for battery energy storage systems to account for carbon impacts alongside financial objectives. However, estimating hourly MEFs remains challenging in interconnected power systems, where cross-border flows affect the marginal generation mix. This paper develops two data-driven estimators for hourly MEFs using consumption-based emissions and net load time series derived from a multi-regional input-output accounting framework. The proposed estimators consist of a Marginal-Mix based method and a regression-based Linear Model Tree, which captures regime-dependent marginal responses using observable system-state variables. The estimators are evaluated for the DE-LU and DK2 bidding zones using data from 2022 to 2025. Compared with the Siler-Evans net-load binning baseline, the Linear Model Tree improves out-of-sample R squared from 0.56 to 0.72 in DK2 and from 0.87 to 0.92 in DE-LU, demonstrating the value of regime-dependent Marginal Emissions Factor estimation in interconnected power systems.**

***Index Terms*—Marginal emissions factor, carbon-aware battery operation, interconnected power systems**

I. Introduction

Grid-level energy storage, particularly battery systems, is increasingly recognized as a cornerstone of the energy transition, enabling higher penetrations of variable renewable energy and enhancing system flexibility and reliability. By absorbing surplus renewable generation and releasing it during periods of scarcity, batteries can support both grid stability and decarbonization objectives. However, in practice, most battery scheduling is driven by price-spread arbitrage, charging when electricity is cheap and discharging when it is expensive, which maximizes revenue but often treats emissions as a secondary concern [1]–[3]. Therefore, batteries can also inadvertently increase emissions if charged during high-carbon periods or if their operation leads to increased fossil generation elsewhere on the grid [1], [2], [4], [5]. Studies have shown that modest adjustments to operational patterns, guided by MEFs, can achieve emission reductions at minimal cost to operators [4]. This underscores the importance of developing accurate and high-resolution MEFs for operational decision-making.

To implement temporal load-shifting toward hours when marginal generation is cleaner, it is essential to quantify how incremental changes in demand affect total system emissions, a relationship captured by the Marginal Emissions Factor (MEF). Unlike Average Emissions Factors (AEFs), which provide an attributional perspective by characterizing the mean emissions intensity of the electricity mix, MEFs adopt a consequential perspective, measuring the change in system emissions induced by a marginal change in demand and reflecting the emissions intensity of generators that actually respond to incremental load changes [1], [2], [6]. This distinction is not merely conceptual: several studies demonstrate that AEFs can substantially misestimate avoided emissions and may even yield misleading operational recommendations when used to guide flexible demand [6], [7]. For batteries, whose charging and discharging decisions are highly flexible, an operationally meaningful MEF is therefore essential for integrating carbon reduction into scheduling objectives [4]. Two broad families of MEF estimation methods exist in the literature. Simulation-based approaches use detailed unit-commitment or economic-dispatch models to compute MEFs and analyze longterm scenarios by simulating the power plant dispatch. [8]–[11]. Though powerful, these dispatch tools can require substantial computational resources, specialized training, licensed software, and proprietary data. These requirements hinder their usability for operation purposes [12]. Statistical approaches, by contrast, infer MEFs directly from historical observations of demand and generation, at the cost of additional structural assumptions about the form of the marginal response. Hawkes [13] and Siler-Evans et al. [14] established the regression-based foundations of the field. Huber et al. [6] extended this framework to the operational context of electric vehicle smart charging, demonstrating that MEF-based scheduling yields significantly greater emissions reductions than AEF-based strategies. Subsequent work has refined both the statistical machinery and the scope of the problem. Zohrabian et al. [15] proposed a multiple linear regression model using historical demand, variable renewable generation, and CO_2 emissions data to quantify hourly MEFs, explicitly incorporating the emissions embedded in net electricity imports alongside in-region generators, expressing MEF only as a function of net

load, hour and month. Peters et al. [16] addressed the marginal mix composition at hourly resolution, though without detailed treatment of cross-border flows. Beyond regression, Ben Amor et al. [8] employed a Markov Switching Dynamic regression model, underscoring the importance of conditioning MEF estimates on the current system state, while Beltrami et al. [17] applied an ARIMA time-series approach to decompose emissions dynamics and [18] presented an MLP-linear composite model to evaluate and forecast MEF. Despite this progress, the validation of MEF models remains an open methodological challenge. No ground-truth MEF observations exist against which models can be directly evaluated. Koebrich et al. [12] recently addressed this gap by proposing a suite of indirect validation tests that assess the plausibility of the MEF model without requiring ground truth. This framework provides a principled basis for comparing MEF estimation methods and identifying their respective failure modes.

Our paper makes three main contributions:

- Integration of cross-border imports into marginal-mix construction via a multi-regional input-output (MRIO) framework, enhancing the accuracy of emissions accounting for interconnected power systems.
- Development of a regression-based MEF estimator using linear model trees, representing an advancement over classic statistical approaches, alongside a comparative analysis with a ratio-based marginal-mix intensity estimator.
- Empirical validation of both estimators across net load, prices, temporal patterns, and generation mix composition, revealing distinct statistical properties while consistently recovering expected marginal emissions trends.

The remainder of the paper is organized as follows. Section II presents the MRIO-based methodology for constructing the Emissions and Net Load time series. Section III describes the two MEF estimation approaches and their identification assumptions. Section IV compares the behavior of the indicators across system conditions and discusses their implications for battery scheduling. The analysis is conducted on the DE-LU and DK2 bidding zones, which exhibit distinct generation and interconnection characteristics. Section V summarizes the findings and outlines directions for future work.

II. MRIO MODEL

Our consumption-based emission methodology builds on the multi-regional input-output framework proposed by Clauß et al. (2018) [19], which computes hourly average CO2eq. intensities at the bidding-zone level while explicitly accounting for cross-border electricity imports instead of treating each zone as isolated. It traces the electricity consumed in a given bidding zone back to its contributing technology sources, including imported electricity, and thus quantifies the share of each technology in local consumption. In practical terms, the method gives a consumption-based representation of the electricity mix at each time step: it captures the emissions associated with electricity use, defines the net load as load minus variable renewable electricity (VRE), including imported VRE from neighboring zones, and provides a more detailed picture of how the bidding-zone demand is supplied across technologies.

TABLE I
GHG EMISSIONS FROM ELECTRICITY GENERATION TECHNOLOGIES - ($kgCO_2eq/MWh$)

Technologies	Values	Technologies	Values
Coal[1]	820	Wind Onshore[1]	11
Gas[1]	490	Wind Offshore[1]	12
Hydropower[1]	24	Solar Phototovoltaics[1]	48
Nuclear[1]	12	Import GB[2]	175
Biomass[1]	230		

[1] IPCC AR 5 (2014) [20] ; [2] AEF Electricy Maps 2024 [21]

A. Bidding zone exploration

The first phase is a bidding zone classification. A distinctive feature of our approach is the dynamic classification of neighboring bidding zones. Instead of assigning zones a priori, we start from the target bidding zone and recursively explore its neighbors. A neighboring zone is classified as *variable* if its share of imports in the study zone's total imports exceeds a threshold ϵ. In that case, its own neighbors are also explored. If the import share falls below the threshold, the zone is treated as a *boundary* zone, meaning that its imports are not further decomposed and it enters the model as an electricity generation technology. The set of variable zones is noted $\mathcal{V}$ while the set of boundary zone is $\mathcal{B}$

This refinement makes the boundary of the model endogenous to the electricity flow structure rather than predefined. It also avoids over-expanding the system into zones whose contribution to the target zone is negligible. Countries without data on ENTSOE, such as Great Britain and Russia, are directly considered as boundary countries.

B. Generators Representation

We further refine the model by not aggregating identical technologies across bidding zones. Instead, we distinguish them by both technology and bidding zone. This yields a more granular representation of the electricity mix and account for any differences in how the technologies perform across bidding zones. Introducing the set of technologies $\mathcal{T}$, generators $\mathcal{G}$ are then defined as: $\mathcal{G} = (\mathcal{T} \cup \mathcal{B}) \times \mathcal{V}$.

C. Emission Factors

We distinguish two types of entries in the model. The first category covers actual generation technologies, for which we assign fixed CO2 intensity factors defined in table I. The second category covers fixed bidding zones, for which the CO2 intensity is computed hourly using only local generation.

D. Calculation process

This section provides an overview of the calculation details. First, we assemble hourly generation data by bidding zone and electricity generation technology. For each variable country, data are organized as a matrix $T(t)$ that includes the electricity

generation from each generation technology and imports from neigbouring zones $g_i \in \mathcal{G}$ for all variable zone $v_i \in \mathcal{V}$.

$$T(t) = \begin{pmatrix} P_{v_1,g_1} & \dots & P_{v_n,g_1} \\ \vdots & & \vdots \\ P_{v_1,g_k} & \dots & P_{v_n,g_k} \end{pmatrix} \tag{1}$$

Second, we normalize each bidding zone's generation to 1 MWh and derive the production-share matrix $N(t)$.

$$N(t) = \begin{pmatrix} \frac{1}{\sum_i P_{v_1,g_i}} & 0 & \dots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \frac{1}{\sum_i P_{v_n,g_i}} \end{pmatrix} \tag{2}$$

We derive the production matrix $P(t) = T(t) \cdot N(t)$, containing the share of electricity from each technology and the import on the total hourly generation in a respective variable zone.

To include the share of each technology in the import we separate $P(t)$ in two parts: (i) Imports from variable zone $P_v(t)$ and (ii) generation from each technology and imports from boundary zone $P_g(t)$.

$$P(t) = \begin{pmatrix} P_v(t) \\ P_g(t) \end{pmatrix} \tag{3}$$

For each bidding-zone, electricity generation and imports from a given generator must be either consumed within the bidding-zone or exported. This follows the logic of the MRIO framework, which is used to estimate consumption-based emissions at the country or regional level [22]. Based on this approach, the electricity balance can be written as:

$$M(t) = P_g(t) + M(t) \cdot P_v(t) \tag{4}$$

$$\implies M(t) = P_g(t) \cdot (I - P_v(t))^{-1} \tag{5}$$

The Mix matrix M represents, for each hour of the considered period, the share of each electricity generation technology and boundary bidding zone in the electricity use and exports of bidding zone j.

For a defined zone j, for every hour of the year t, the AEF, consumption-based emissions E, net load L_{net} and the consumption of each generator types, are calculated by:

$$AEF_{jt} = \sum_{g \in \mathcal{G}} \zeta_g M_{(g,j)}(t) \tag{6}$$

$$E_{j,t} = AEF_{j,t} \cdot L_{j,t} \tag{7}$$

$$L_{net,j,t} = \sum_{g \in \mathcal{G} \setminus \text{VRE generators}} L_t M_{(g,j)}(t) \tag{8}$$

$$G_{j,g,t} = M_{(g,j)} \cdot L_{j,t} \tag{9}$$

where t is the hour of the year, j is the index of a specific bidding zone, $\mathcal{T}$ is the set of electricity generation technology, including boundary zones, ζ is a vector containing the emission factors of the technology and fixed zones and finally L_j is the load of the considered zone.

III. MEF ESTIMATION

A. Marginal Mix

For our application, we focus on the technologies that are likely to respond at the margin to demand changes. Following Peters et al. [16], the short-term marginal mix isolates controllable generation that adjusts when net load changes, while excluding technologies whose variation is exogenous or moves counter to the net-load signal. This matters because an observed increase in generation is not necessarily induced by demand: for instance, hydro may ramp up when fossil output falls, or imports may fluctuate for reasons unrelated to local demand. In such cases, a marginal emissions factor based on the raw generation mix would misrepresent the technology actually supplying the incremental load. A marginal-mix preprocessing step is therefore needed to filter out technologies whose hourly changes are not driven by net load and to retain only those that are plausibly displaced by an additional unit of demand. In the DK2 zone, gas and coal generation move in the opposite direction to net load in 37% of the hours in our sample (2022-2025), suggesting that a non-trivial part of their variation is not captured by a naive net-load-based attribution. In SE4, hydro and imported hydro from SE1 move opposite to net load in 35% and 31% of hours, respectively, which again shows that generation changes are often driven by balancing or exogenous conditions rather than by demand alone.

Marginal mix generation and marginal mix emissions are thus respectively computed hourly as:

$$\Delta\tilde{G}_t = \sum_{i \in \{g \in \mathcal{G}, \Delta G_{gt} \Delta NL_t \geq 0\}} \Delta G_{it} \tag{10}$$

$$\Delta\tilde{E}_t = \sum_{i \in \{g \in \mathcal{G}, \Delta G_{gt} \Delta NL_t \geq 0\}} \Delta G_{it} \zeta_i \tag{11}$$

While the sign filter excludes a fraction of observations on a per-generator, per-hour basis, it does not introduce systematic bias in expectation: excluded hours are those in which observed generation changes are inconsistent with net-load-driven dispatch, and their removal improves the signal-to-noise ratio of the marginal mix without selectively favouring any particular regime or technology. Both the regression-based model and the marginal-mix based MEF rely on a common construction of a marginal mix that isolates generation changes consistent with net-load-driven dispatch.

B. Marginal-Mix based MEF

Using those calculated time series, we compute the estimated MEFs through the following equation:

$$MEF_t = \frac{\Delta\tilde{E}_t}{\Delta\tilde{G}_t}, \tag{12}$$

Importantly, this estimator should not be interpreted as a regression slope or a finite-difference derivative. It corresponds to a CO_2-intensity-weighted average of the marginal mix composition, where weights reflect the relative contribution of each technology to net-load driven generation changes. Unlike the regression-based specification, this estimator provides a

pointwise measure of emissions intensity without pooling observations across multiple hours or load regimes. It is therefore easy to interpret and computationally simple, but it is also more sensitive to measurement noise and filtering bias.

C. Regression based MEF

To analyse trends in the marginal emission factor (MEF), Siler-Evans et al. [14] estimate separate regressions on subsets of the data defined by a partitioning variable. Load and net load are often used for this purpose. This approach, however, implicitly assumes that the chosen partitioning variable is the primary determinant of the MEF. While net load is plausibly related to the MEF through merit-order dispatch, it may not capture all relevant drivers of marginal generation.

We instead partition the data with a decision tree and estimate a linear model in each terminal node to assess the quality of the split. This yields an interpretable model that retains the regression-based structure commonly used in the literature. Such models are known as linear model trees [23]. Fig. 1 illustrates the methodology used.

The linear model tree is fitted to the marginal mix series $(\Delta\tilde{G}_t, \Delta\tilde{E}_t)$. Let Z denote the set of splitting variables. For each terminal node i, we estimate

$$\Delta\tilde{E}_t = \beta_i \Delta\tilde{G}_t + \epsilon_t \tag{13}$$

The leaf-specific slope β_i is obtained by OLS linear regression and interpreted as the MEF. The model do not include an intercept as, by construction, data are expected to be centered.

The split variables include zone-level electricity demand, day-ahead prices in the zone and adjacent zones, interzonal price differentials, total generation in the zone, imports and temporal indicators such as hour, month, and year. Price differentials serve as a proxy for cross-border dispatch incentives. These variables are used only to build the tree structure, at deployment, they are used solely to route each observation to a leaf and then apply the corresponding MEF. We select these variables to represent state of the system and for exploratory analysis, with the objective of identifying which factors are most influential in determining the MEF level.

For deployment, these variables would need to be forecasted. However, this is outside the scope of the present analysis so as not to introduce errors from the forecast uncertainty.

Model development follows a structured procedure. The hyperparameters, tree max depth and the minimum number of sample in each leaf, are first selected by cross-validation. The model is then trained using the selected settings. Feature selection is subsequently performed using feature importance, after which the model is re-constructed on the reduced feature set.

IV. Results

A. Linear Model Tree

1) Comparison with Siler-Evans: To assess model performance, we generated data from 2022 to 2025, using the last year as the test set and the remaining years as the training set. As a baseline for evaluating the Linear Model Tree approach, we used the standard linear regression and the Siler-Evans methodology with net-load binning. Table II reports the coefficient of determination R^2 and mean absolute percentage error (MAPE) for both methods. The Linear Model Tree outperforms both baselines, with the largest improvement observed in DK2, where test R^2 increases from 0.60 to 0.72 and MAPE reduces by 28 %. This indicates that the tree-based model is better able to describe how the delta in emissions from the marginal mix changes across different operating regimes. In contrast, DE-LU shows a more modest but still meaningful improvement, indicating that the baseline already performs well in that zone.

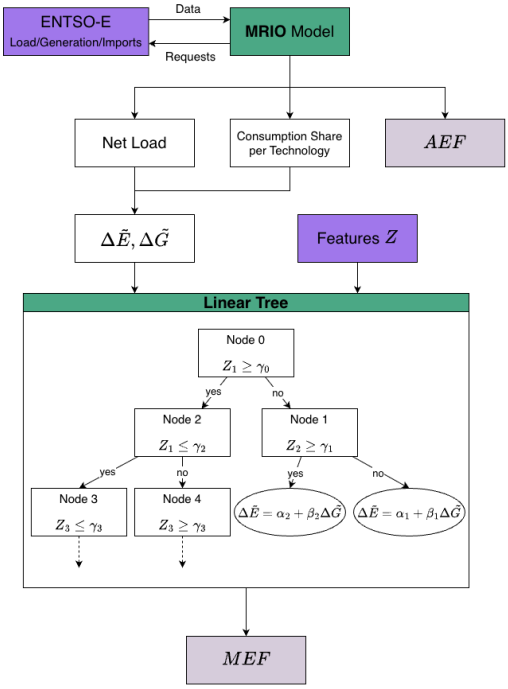


Fig. 1. Linear Model Tree methodology

TABLE II
Coefficient of determination R^2 and MAPE (%) for Siler-Evans and Linear Model Tree method

	Linear Regression		Siler-Evans		Linear Model Tree	
Zones	***Train***	***Test***	***Train***	***Test***	***Train***	***Test***
DK2	0.61	0.60	0.57	0.56	0.77	0.72
	123	118	110	102	92	90
DE-LU	0.85	0.85	0.87	0.87	0.93	0.92
	73	75	70	73	57	59

2) Features importances: The importance of variable z_j is defined as the cumulative reduction in mean squared error (MSE) of the linear model in the leaves induced by all splits on z_j. Hence, the more a feature contributes to partitioning the data into homogeneous regions, the higher its importance.

Table III shows top 5 features according to feature importance and highlights clear differences for the two zones. In DE-

TABLE III
FEATURES IMPORTANCE FOR DE-LU AND DK2 BIDDING ZONES

DE-LU		DK2	
Feature	***Importance***	***Feature***	***Importance***
$AEF * L/L_{net}$	0.16	ΔI_{SE4}	0.35
L_{net}	0.12	Hour	0.11
Hour	0.11	AEF	0.11
ΔI_{CH}	0.08	Price_{DE-LU}	0.09
Price_{DE-LU}	0.07	$AEF * L/L_{net}$	0.06

I= Import, G= Total local generation

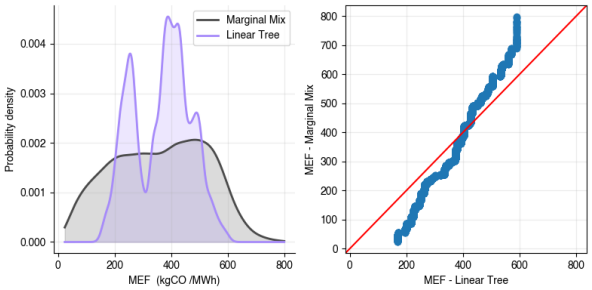


Fig. 2. DE-LU - Kernel Density Estimates plot and Q-Q plot for both Marginal Mix and Linear Tree methods

LU, net load and the time of day are among the most influential variables. By contrast, in DK2, the most important features for explaining the MEF are linked to neighboring bidding zones, especially the variation of imports from SE4. AEF and hour are also used by the model to split the data.

B. MEF trends

While ground-truth Marginal Emissions Factors (MEFs) cannot be directly observed, Koebrich et al. (2025) [12] demonstrate that their accuracy can be indirectly assessed by evaluating whether they exhibit expected behavioral patterns under varying grid conditions. Following this methodology, we analyze the performance of our linear tree MEF and direct marginal mix MEF estimators for the DE-LU and DK2 bidding zones in 2025. Data were grouped by net-load level, day-ahead (DA) price, share of coal in the consumption mix, and hour of day, with mean values calculated to quantify whether our results align with theoretical expectations.

1) DE-LU Analysis: The Kernel Density Estimate plot on Fig. 2 reveals distinct distributional characteristics for the Marginal Mix and Linear Tree MEF estimators, both constrained within the 0-850 $kgCO_2eq/MWh$ range, corresponding to coal's carbon intensity. The Linear Tree estimator exhibits pronounced spikes at specific values. In contrast, the Marginal Mix estimator displays a flatter, more continuous distribution, reflecting its greater sensitivity to the full spectrum of marginal conditions and less reliance on discrete fuel-type thresholds. The Q-Q plot on the right illustrates distributional similarity. While the majority of values fall along the red diagonal line indicating broad agreement, the Marginal Mix estimator shows greater dispersion at the tails, capturing a wider range of marginal behaviors and extreme

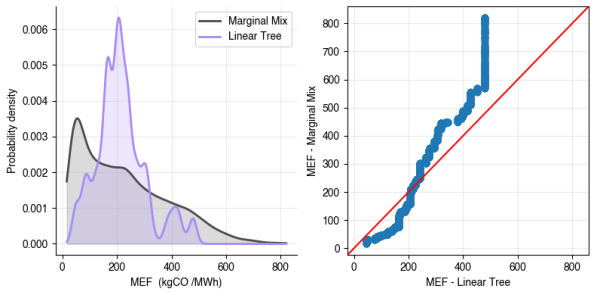


Fig. 3. DK2 - Kernel Density Estimates plot and Q-Q plot for both Marginal Mix and Linear Tree methods

system conditions. This suggests that while both estimators align in central tendencies, the Marginal Mix approach is more sensitive to nuanced system variations. From Fig. 4, both MEF estimators exhibit qualitatively similar trends but differ in magnitude and sensitivity to system conditions. The regression-based MEF tends to be smoother with a narrower interquartile range (IQR), suggesting greater robustness to noise in the data. A clear diurnal variation emerges in the hourly plot, with higher MEFs at night and lower MEFs during peak hours. This aligns with Koebrich et al.'s Expected Net-Demand Temporal Patterns test, which posits that MEFs should be higher during off-peak periods in coal-heavy regions due to coal's operational constraints and high carbon intensity. Similarly, the net-load plot reveals that MEFs increase as net load decreases, consistent with the expectation that coal, with its slower ramping capabilities, remains marginal during low-demand periods. Conversely, higher net loads correlate with lower MEFs, as more flexible gas plants are dispatched to meet peak demand. A paired hour-by-hour comparison yields a Spearman rank correlation of $\rho = 0.55$, indicating moderate agreement on the relative ordering of emissions intensity across hours. While both models capture the same broad seasonal and diurnal patterns, they diverge on the precise ranking of individual hours, particularly at the extremes, as evidenced by the upper-tail compression observed in the Q-Q plot.

2) DK2 Analysis: Fig. 3 presents the marginal distributions and Q-Q comparison of the two MEF estimators for the DK2 bidding zone. The KDE reveals a fundamental difference in distributional shape: the Linear Tree produces a concentrated distribution peaking near 150 $kgCO_2eq/MWh$ with limited spread, while the Marginal Mix yields a broad right-skewed distribution spanning the full 0-850 $kgCO_2eq/MWh$ range.The Q-Q plot corroborates these observations. The two estimators align closely along the diagonal in the 0-250 $kgCO_2eq/MWh$ range, indicating agreement in the hydro-dominated regime. Above 250 $kgCO_2eq/MWh$, the Marginal Mix diverges sharply upward, reaching values exceeding 800 $kgCO_2eq/MWh$ while the Linear Tree remains bounded near 450 $kgCO_2eq/MWh$. A Spearman rank correlation of $\rho = 0.45$ confirms moderate temporal co-movement: both models identify the same broad emissions regime at any given hour, but the observed

TABLE IV
SUMMARY OF KOEBRICH ET AL. [12] TEST RESUTS

Description of test	**DE-LU**		**DK2**	
	MMR	*LMT*	*MMR*	*LMT*
MEFs should not be negative	P	P	P	P
MEFs should not exceed the most polluting generator	P	P	P	P
MEFs should be correlated with the percentage of coal in the fuel mix	P	P	P	P
MEFs during peak net-demand periods are expected to be lower than in low net-demand periods in regions with coal generation	P	P	-	-
MEFs during peak net-demand periods are expected to be higher than in low net-demand periods in regions without coal generation	-	-	P	P
MEF models are expected to contain information about real-time curtailment to be useful for device optimization	F	F	F	F

MMR: Marginal-Mix MEF; LMT: Linear Model Tree; P: Pass; F: Fail

divergence suggests that operational decisions based on the two indicators could differ under certain system conditions. Fig. 5 illustrates the behavior of the MEF estimators in DK2, where more similar trend between MEF series and AEF is observed. Despite this, both MEF indicators exhibit similar trends when analyzed against system conditions for all key variables. The hour-of-day plot shows no clear diurnal pattern, with a wide IQR across all hours. This lack of hourly periodicity may reflect DK2's high renewable penetration and interconnection with neighboring zones, which can mask typical diurnal marginal patterns. However, the share-of-coal plot reveals a sharp increase in MEFs with rising coal share for both estimators.

3) Summary: Table IV summarizes the results from Koebrich et al. [12] tests, demonstrating that both MEF estimators exhibit the expected behavior across all tests except for curtailment information, which is not directly included in our estimators. These results underscore the relevance of our methodology and indicate that the choice for operational purposes should be based on the ability to use these signals during operation.

V. CONCLUSION

This work contributes a consumption-based MRIO framework for interconnected grids and introduces a Linear Model Tree MEF estimator that improves on conventional net-load-binning approaches by capturing regime-dependent dispatch behavior more effectively. The framework combines a consumption-based emissions model with two statistical MEF estimators: a direct marginal-mix ratio and a Linear Model Tree conditioned on observable system-state variables.

The comparative evaluation of the two estimators yields several actionable findings. First, the proposed Linear Model Tree outperforms the Siler-Evans net-load binning baseline. This confirms that conditioning the marginal response on a richer set of system-state variables, including cross-border import levels, day-ahead prices, and temporal indicators, captures regime-dependent dispatch behaviour that a single partitioning variable cannot reproduce. Both methods are validated against the four indirect tests proposed by Koebrich et al. [12], and both exhibit the expected behavioural patterns with respect to net load, coal share, and hour of day, providing confidence in their physical plausibility despite the absence of ground-truth MEF observations. The Linear Tree offers greater robustness and smoother behaviour across typical operating conditions, making it more suitable for deployment in a forecasting pipeline. The Marginal Mix, by contrast, provides a more informative signal at the high-emission extremes that are most consequential for emissions reduction, at the cost of higher sensitivity to measurement noise.

Several limitations of the present work point to directions for future research. The operational deployment of the Linear Model Tree requires forecasts of the splitting variables at the decision horizon. Integrating the MEF estimation with a short-term forecasting module and quantifying the propagation of forecast uncertainty into the MEF signal remains an open problem. Finally, applying the framework to battery dispatch optimisation with the estimated MEF in the scheduling objective, would allow the emissions impact of the two estimators to be evaluated directly in terms of avoided CO_2, closing the loop between emissions modelling and operational decision-making.

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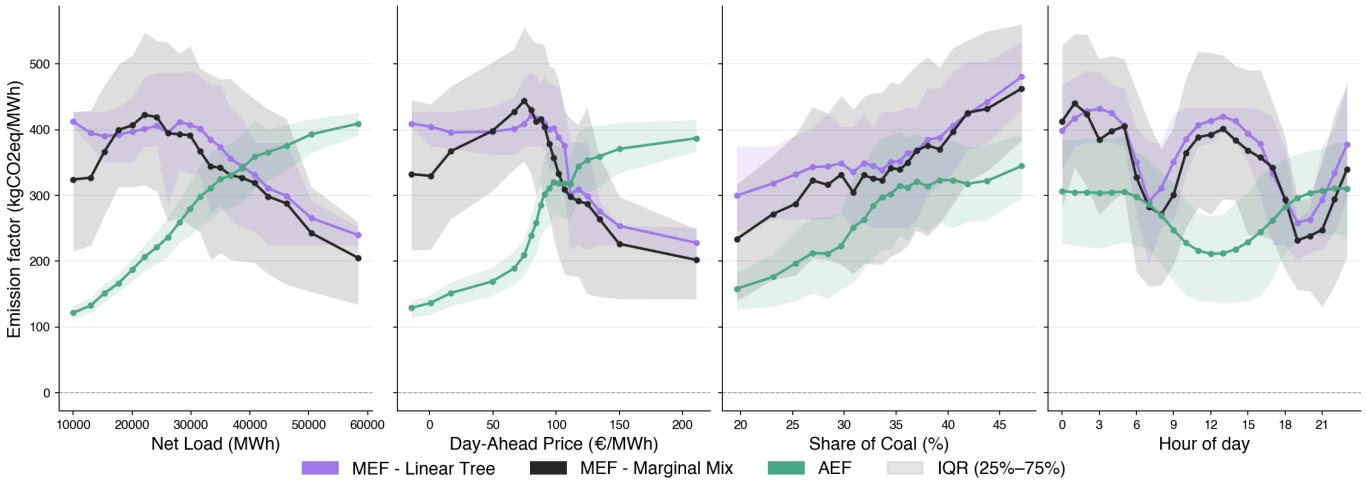


Fig. 4. DE-LU - MEF and AEF trends with Net Load, Day-Ahead Price, Share of Coal and Hour of the day

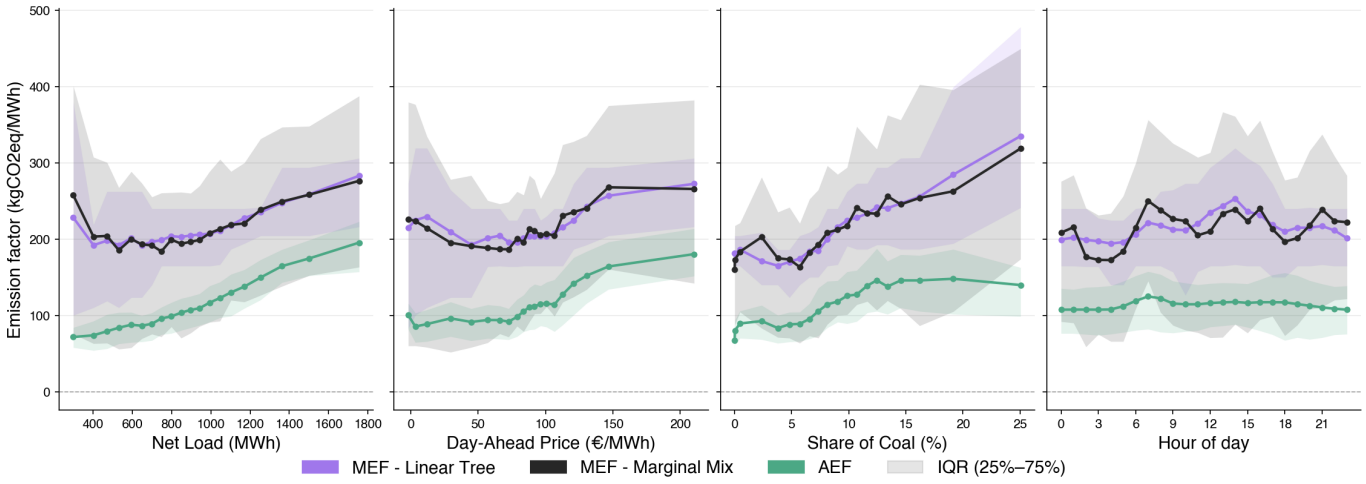


Fig. 5. DK2 - MEF and AEF trends with Net Load, Day-Ahead Price, Share of Coal and Hour of the day

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